

SWE40006

Software Deployment and Evolution

Lecture 7

Intro to Project
Review for Test

Lecture overview

This lecture will cover,

- Intro to Project (see Project Spec)
- Review for Test

Project

1. DevOps pipeline
2. Deployment environments & servers
3. DevOps tool chain
4. A software application
5. Different (4) levels of attempt/tasks
6. Assessment Requirements: 3 stages/submissions (weeks 8, 11, and 12)
7. Group project:
 - a. Group work and group submissions; no individual submissions
 - b. Equal contribution expected (quantity/quality); or marks adjusted
8. See Project Description and 3 Assignment Specs

Review for Test

Contents

1. Lectures (weeks 1~8)
2. Tutorials (weeks 1~8)
3. Quizzes (weeks 1~8)
4. Readings ...
5. “Deployment Tasks (1~4)” are helpful

Test format

- a. In week 9 tutorial – in person only
- b. Closed book
- c. 90 minutes: Arrive on time; (*EAP considerations/arrangements (email JH) with EAP*)
- d. ~20 questions; multiple choice and short answer
- e. Sequential – one question at a time; can review before submission

Content Highlights

1. Software Deployment
2. Software Evolution (Maintenance)
3. The software application: lifecycle activities ...
4. Software stages/teams: dev, test/QA, Operations/support
5. Software environments: Development, testing, staging, production/live
6. Tool support/chains & automation: ...
7. DevOps, CI/CD, CD, Infrastructure as Code, ...

Software Deployment in a Nutshell

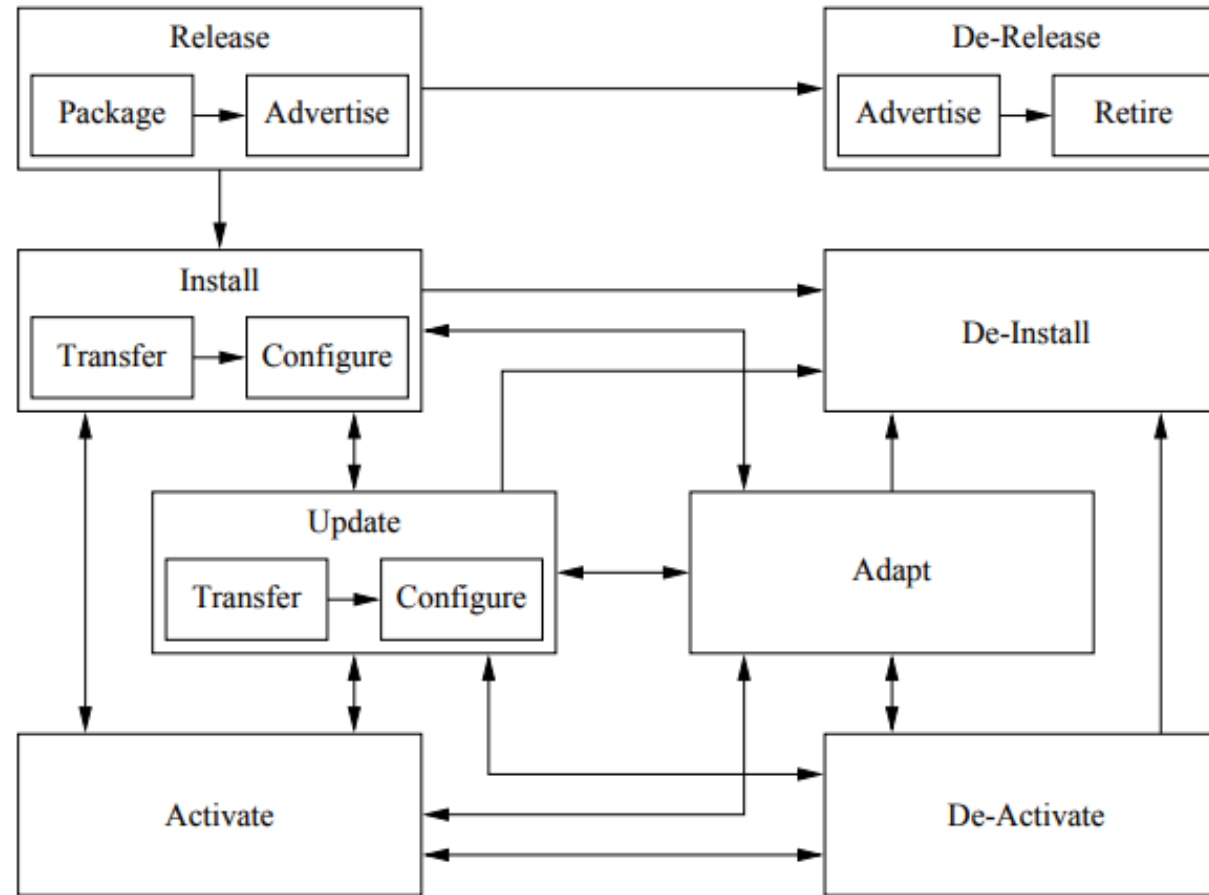
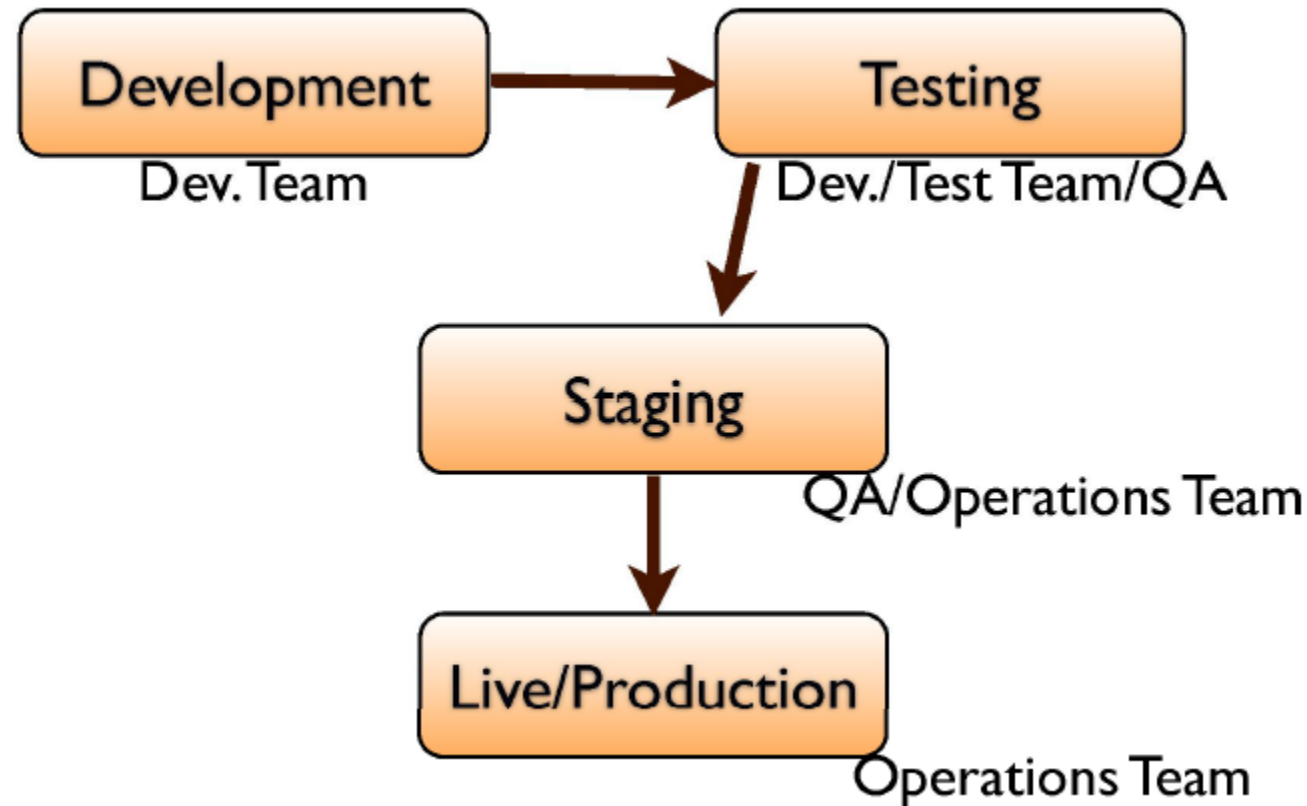


Figure 1: Activities of the Software Deployment Process.

<https://www.ics.uci.edu/~andre/papers/T3.pdf>

Different Types of Environments



- Software is released for different target environments
- Contents of software can change depending on environment
- Teams that focus on different environments/infrastructures

Software Deployment Approaches: Manual vs. Automated

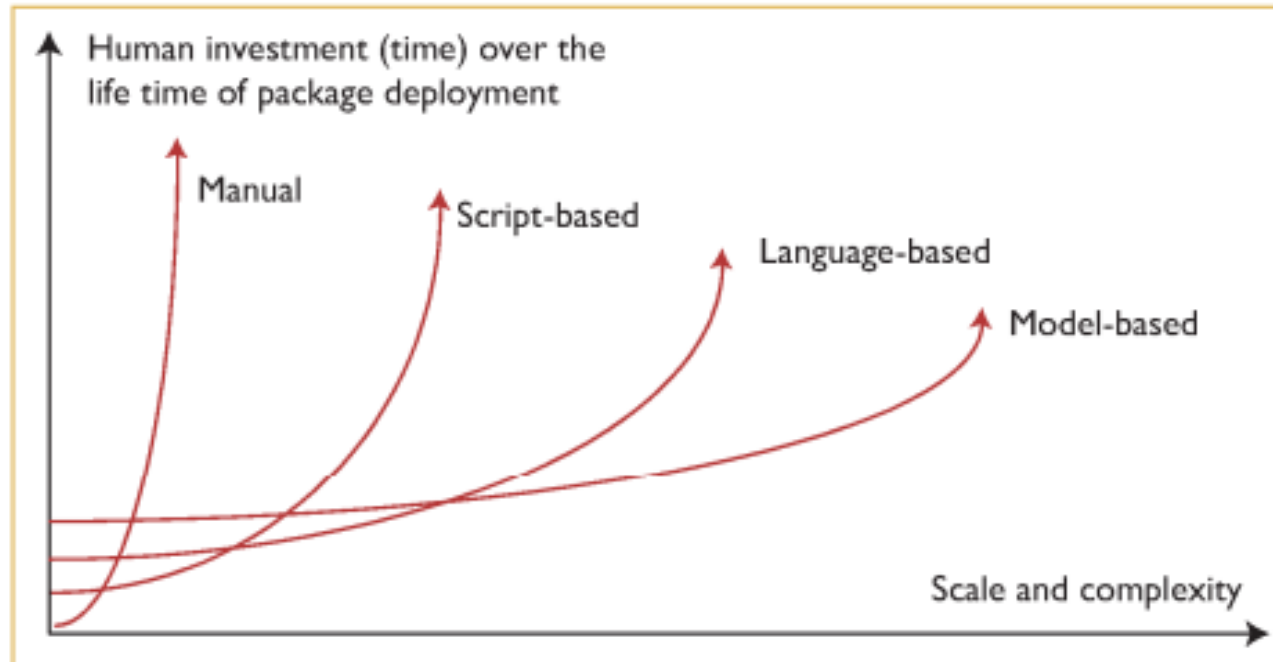


Figure 1. Trade-offs between initial cost (development and learning) and repeated use cost. The level of a tool's automation pushes costs up earlier in the development cycle, but developing tools, learning from them, and creating templates with them pays off as complexity increases.

Software Deployment – Planning Framework & Data Migration

- The following information is needed:
 - Deployment Model (for scope of work)
 - Constraints (all the below are inter-related)
 - Temporal
 - Environment and Infrastructure
 - Automation
 - Communication
 - Change pattern/properties
 - Impact of deployment & Monitoring
- Data migration ...

Software Evolution in SDLC

1. SDLC models ... software change & evolution
2. Types of software changes:
corrective, perfective, preventative, inventive
3. Major software changes
 - a. Refactoring
 - b. Re-structuring
 - c. Re-engineering
4. Evolution of software deployment (v evolution of software)

DevOps - Evolution of Software Deployment(/Development)

1. CI/CD (continuous integration/continuous delivery)
2. DevOps
3. IaC, CD (continuous deployment)
4. DevSecOps
5. DevOps framework (CALMS), activities (PCBT & RDOM), and processes ...
6. DevOps tools and tool chains - automation

DevOps processes

- Source code management
- Source code review
- Configuration management
- Build management
- Repository management
- Test automation
- Release management
- Continuous integration
- Continuous delivery
- Continuous deployment

Q & A

- Any questions?